HW #6 — Continuous Random Variables (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Monday, 11/2 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as one .pdf files.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

We will now focus on continuous random variables, using refined methods for finding items such as PDFs, CDFs, expected values, and variances.

Note: This homework will involve a lot of integration. To that end, please look over the following methods:

- Inetgration by parts
- Chain rule
- U-substitution
- L'Hopital's rule

Textbook Problems (4-5 pts each)

From *Introduction to Probability, Second Edition*:

- Chapter 5: Q1, 5, 8, 9, 36, 48

Please note the following for these problems:

- For Q5(a): $E(X^4) = \frac{1}{5}$ for $X \sim Unif(0, 1)$
- For Q36: Recall the memoryless property, which is seen in Definition 5.5.2 in the textbook.